

ISTQB® CTFL v4.0

Chapter 6

Test Tools

Main reference: ISTQB CTFL syllabus. Speaker notes expanded with Stapp textbook explanations.



Chapter 6 learning route

Syllabus numbering retained: 6.1 → 6.2

6



6.1 Tool Support for Testing

FL-6.1.1 (K2): Explain how different types of test tools support testing.

6.2 Benefits and Risks of Test Automation

FL-6.2.1 (K1): Recall the benefits and risks of test automation.



INTERACTIVE: Ask learners which tools they already use in testing.

6.1 Testing activities that tools can support

Tools can assist throughout the testing process.



1 Product analysis and evaluation

2 Determining measures used in testing

3 Test analysis

4 Test design

5 Test implementation

6 Test execution

7 Analyzing test results

8 Test environment management



INTERACTIVE: Which of these activities have you already supported using a tool?

6.1 Why use test tools?

The purpose is to improve efficiency, reliability, consistency, and capability.

Automate repetitive or resource-intensive work

Examples: regression tests, repeated data entry, expected-versus-actual comparison.

This can increase testing efficiency.

Support manual testing activities

Tools can make manual activities more efficient.

Better support can improve the reliability of testing.

Increase consistency and reproducibility

Repeatable execution helps reproduce defects and improves testing quality.

Enable difficult testing

Some activities, such as performance testing, may be very difficult or impossible to perform manually.



INTERACTIVE: Which purpose is most important for regression testing? What about performance testing?

6.1 Tool support for testing

Different tools support different testing activities.

Management tools

- Application lifecycle management
- Traceability from test basis to requirements
- Test monitoring, defect management and configuration management
- Team-work support and automated reporting

Static testing tools

- Support reviews
- Support static analysis

Test design & implementation tools

- Facilitate generation of test cases, test data and test procedures

Test execution & coverage tools

- Facilitate automated test execution and coverage measurement



INTERACTIVE: Ask learners to map tools they know to each category.

6.1 More tool categories

Tool support extends beyond functional test execution.

Non-functional testing tools

Support testing that is difficult or impossible to perform manually.

DevOps tools

Support delivery pipelines, workflow tracking, automated builds and CI/CD.

Collaboration tools

Facilitate communication among people involved in testing.

Scalability & deployment tools

Examples include virtual machines and containerization tools.

Any other tool that assists in testing can be a test tool in context — even a spreadsheet.



INTERACTIVE: Which tools provide the fastest feedback in a CI/CD workflow?

6.2 Benefits of test automation

Automation can save time, improve repeatability and provide better information.

Time saved

Reduce repetitive manual work such as regression execution, repeated data entry and result comparison.

Consistency

Prevent simple human errors through greater consistency and repeatability.

Objective assessment

Provide measures such as coverage that may be too complicated to derive manually.

Better information

Provide statistics, graphs and aggregated information for management and reporting.

Faster feedback

Reduce execution time, support earlier defect detection and faster time to market.

Deeper testing

Free testers to design new, deeper and more effective tests.



INTERACTIVE: Which repetitive task would be the best automation candidate in your current context?

6.2 Risks of test automation

Tool adoption needs realistic expectations and appropriate governance.



Unrealistic expectations

Overestimating functionality, ease of use or expected benefits.

Underestimated effort

Misjudging time, cost and effort for introduction, maintenance and process change.

Wrong use case

Using automation where manual testing is more appropriate, such as human usability evaluation.

Over-reliance

Ignoring the continuing need for human critical thinking.

Vendor / open-source dependency

Vendor failure, poor support, retirement, or abandonment of an open-source tool.

Compatibility / compliance

Platform incompatibility or failure to meet regulatory or safety requirements.



INTERACTIVE: What is the danger of trusting an automated test suite blindly?

6.2 Automation or manual testing?

Choose the approach that best fits the testing objective.



Good automation candidates

- Regression tests executed repeatedly
- Re-entering the same test data
- Expected-versus-actual comparison
- Coverage measurement
- Performance testing at scale

Often needs human judgment

- Interface usability evaluation
- Exploratory investigation
- Interpreting unexpected behavior
- Assessing whether results make sense in context
- Deciding what deeper tests to design



INTERACTIVE: For each example, explain why automation or human judgment is more suitable.

Chapter 6 recap and checkpoint

Tools support testing, but testers remain responsible for thinking.



Checkpoint questions

- How do management tools support testing?
- How do execution and coverage tools support testing?
- Which automation benefits relate to speed, consistency and reporting?
- Which risks relate to unrealistic expectations, maintenance and over-reliance?
- When might manual testing be more appropriate than automation?



INTERACTIVE: Ask learners to propose one automation candidate and one test that should remain manual.